Project Support Session
EasyVisa

Agenda

- Project Problem Statement and Dataset Overview
- Open Q&A
- Project Evaluation Rubric
- Project Submission Guidelines
- FAQs

EasyVisa - Business Context

Business communities in the United States are facing high demand for human resources, but one of the constant challenges is identifying and attracting the right talent, which is perhaps the most important element in remaining competitive. Companies in the United States look for hard-working, talented, and qualified individuals both locally as well as abroad.

The Immigration and Nationality Act (INA) of the US permits foreign workers to come to the United States to work on either a temporary or permanent basis. The act also protects US workers against adverse impacts on their wages or working conditions by ensuring US employers' compliance with statutory requirements when they hire foreign workers to fill workforce shortages. The immigration programs are administered by the Office of Foreign Labor Certification (OFLC).

OFLC processes job certification applications for employers seeking to bring foreign workers into the United States and grants certifications in those cases where employers can demonstrate that there are not sufficient US workers available to perform the work at wages that meet or exceed the wage paid for the occupation in the area of intended employment.

EasyVisa - Objective

In FY 2016, the OFLC processed 775,979 employer applications for 1,699,957 positions for temporary and permanent labor certifications. This was a nine percent increase in the overall number of processed applications from the previous year. The process of reviewing every case is becoming a tedious task as the number of applicants is increasing every year.

The increasing number of applicants every year calls for a Machine Learning based solution that can help in shortlisting the candidates having higher chances of VISA approval. OFLC has hired the firm EasyVisa for data-driven solutions. You as a data scientist at EasyVisa have to analyze the data provided and, with the help of a classification model:

- Facilitate the process of visa approvals.
- Recommend a suitable profile for the applicants for whom the visa should be certified or denied based on the drivers that significantly influence the case status.

EasyVisa - Data Dictionary

The data contains the different attributes of the employee and the employer.

Data	Data Description
case_id	ID of each visa application
continent	Information about the continent of the employee
education_of_employee	Information of education of the employee
has_job_experience	Does the employee have any job experience? Y= Yes; N = No
requires_job_training	Does the employee require any job training? Y = Yes; N = No
no_of_employees	Number of employees in the employer's company

EasyVisa - Data Dictionary

The data contains the different attributes of the employee and the employer.

Data	Data Description
yr_of_estab	Year in which the employer's company was established
region_of_employment	Information of foreign worker's intended region of employment in the US.
prevailing_wage	Average wage paid to similarly employed workers in a specific occupation in the area of intended employment. The purpose of the prevailing wage is to ensure that the foreign worker is not underpaid compared to other workers offering the same or similar service in the same area of employment.
unit_of_wage	Unit of prevailing wage. Values include Hourly, Weekly, Monthly, and Yearly.
full_time_position	Is the position of work full-time? Y = Full-Time Position; N = Part-Time Position
case_status	Flag indicating if the Visa was certified or denied

EasyVisa - Q&A

EasyVisa - Evaluation Rubric

Section	Points
Exploratory Data Analysis	8
Data Pre-processing	5
Model Building - Original Data	6
Model Building - Oversampled Data	6
Model Building - Undersampled Data	6
Hyperparameter Tuning	10
Model Performances	5
Actionable Insights & Recommendations	6
Presentation / Notebook - Overall quality	8

EasyVisa - Evaluation Rubric

Section 1: Exploratory Data Analysis

What to Cover?	Guiding Questions and Considerations
Problem definition	What is the core business problem we are trying to solve?
Univariate analysis	What is the distribution of each key variable (normal, skewed, etc.)?
Bivariate analysis	Are there important relationships between key variables?

EasyVisa - Evaluation Rubric

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Section 1: Exploratory Data Analysis

What to Cover?	Guiding Questions and Considerations
Multivariate Analysis	Are there any strong correlations between features?
Use appropriate visualizations to identify the patterns and insights	Are the chosen plots appropriate for the data type? (e.g., Boxplots for wage outliers, etc). Are the plots readable (labels, titles, legends)?
Key meaningful observations on individual variables and the relationship between variables	What are the relationships between the target and other variables? What are some important inter-variable relationships?

EasyVisa - Evaluation Rubric

Section 2: Data Pre-processing

What to Cover?	Guiding Questions and Considerations
Missing Value Detection and Treatment (if needed, with rationale)	Are there null values in the dataset? Should we impute them (mean/median/mode) or drop them?
Outlier Detection and Treatment (if needed, with rationale)	How will you handle outliers in the data? Should we treat them (log transformation/capping) or leave them to preserve information?

Perform each step only if needed, and clearly explain your reasoning, including why you did or did not apply it

EasyVisa - Evaluation Rubric

Section 2: Data Pre-processing

What to Cover?	Guiding Questions and Considerations
Feature Engineering (if needed, with rationale)	Do any variables need transformation? If so, can we extract new features or simplify existing ones?
Prepare data for modeling	Are there any duplicates or inconsistencies (e.g., negative values in certain columns) that were identified and fixed? How should categorical variables be encoded? Have we separated the target variable from the input data? Have we split the data into training and testing sets?

Perform each step only if needed, and clearly explain your reasoning, including why you did or did not apply it

EasyVisa - Evaluation Rubric

Section 3: Model Building - Original Data

What to Cover?

Build at least 5 classification models (Using decision trees, random forest, bagging classifier and boosting methods)

Guiding Questions and Considerations

Have we trained all required models (Decision Tree, Random Forest, Bagging, AdaBoost, Gradient Boosting/XGBoost) on the imbalanced training set?

Which metric is most important for this business problem (Recall, Precision, or F1-Score, etc.)?

You may skip building XGBoost if you face installation issues, but you must still build at least five models in total.

EasyVisa - Evaluation Rubric

Section 4: Model Building - Oversampled Data

What to Cover?

Build at least 5 classification models using oversampled train data (Using decision trees, random forest, bagging classifier and boosting methods)

Guiding Questions and Considerations

Which technique is used for oversampling(e.g., SMOTE)?

Have we trained all required models (Decision Tree, Random Forest, Bagging, AdaBoost, Gradient Boosting/XGBoost) on the Oversampled training set?

Check the model performances and add observations.

You may skip building XGBoost if you face installation issues, but you must still build at least five models in total.

EasyVisa - Evaluation Rubric

Section 5: Model Building - Undersampled Data

What to Cover?

Build at least 5 classification models using undersampled train data (Using decision trees, random forest, bagging classifier and boosting methods)

Guiding Questions and Considerations

Which technique is used for undersampling?

Have we trained all required models (Decision Tree, Random Forest, Bagging, AdaBoost, Gradient Boosting/XGBoost) on the Undersampled training set?

Check the model performances and add observations.

You may skip building XGBoost if you face installation issues, but you must still build at least five models in total.

EasyVisa - Evaluation Rubric

Section 6: Hyperparameter Tuning

What to Cover?	Guiding Questions and Considerations
Choose at least 3 best performing models among all the models built previously (Mention the reason for the choices made)	Which 3 models showed the best potential from the previous steps (Original, Over, or Undersampled)? Give rationale for your selection.
Tune the chosen models	Which hyperparameters are being tuned? Are we using GridSearch or RandomSearch?

EasyVisa - Evaluation Rubric

Section 6: Hyperparameter Tuning

What to Cover?	**Guiding Questions and Considerations**
Check the performance of the tuned models	Did the primary metric (e.g., F1 Score) improve after tuning compared to the base model? Did the tuning increase/reduce overfitting?

EasyVisa - Evaluation Rubric

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Section 7: Model Performances

What to Cover?	Guiding Questions and Considerations
Compare performances of the tuned models and choose a final model.	Comparing all tuned models, which one offers the best trade-off for the business case?
Check the performance of final model on test data.	How does the final best model perform on the unseen Test set? Is the performance consistent with the training/validation results?

EasyVisa - Evaluation Rubric

Section 8: Actionable Insights & Recommendations

What to Cover?	Guiding Questions and Considerations
Conclude with the key takeaways for the business	Add observations and insights based on EDA and model performance on various metrics.
What recommendations would you give to company, based on your analysis, to improve their business?	Based on the analysis, what specific advice can EasyVisa give to OFLC?

EasyVisa - Evaluation Rubric

Section 9: Presentation / Notebook - Overall quality

Low-code Considerations	Full-code Considerations
Structure and flow	Structure and flow
Crispness - Visual appeal	Well commented code
Conclusion and Business Recommendations	Conclusion and Business Recommendations

EasyVisa - Low-Code Version

For learners who aspire to be in managerial roles in the future, focusing on solution review, interpretation, recommendations, and communication with business stakeholders

Download the dataset and the *Learner Notebook - Low Code* (this is a template notebook)

Fill in the blanks in the notebook to complete and execute the code to solve the questions and perform all the tasks as per the grading rubric

Once the notebook is completely executed and necessary outputs obtained, a business presentation (using Microsoft PowerPoint, Google Slides, etc.) has to be created

EasyVisa - Low-Code Version

The presentation should contain observations, insights, and recommendations for the business problem

The presentation template provided can be referred to as a sample

Once the presentation is complete, convert the presentation to .pdf format

The presentation should be submitted as a PDF file (.pdf) and NOT as a .pptx file

Please make sure that all the sections mentioned in the grading rubric have been covered in the submission

EasyVisa - Full-Code Version

For learners who aspire to be in hands-on coding roles in the future, focusing on building solution codes from scratch

Download the dataset and the *Learner Notebook - Full Code* (this is a template notebook containing high-level steps to perform and insight-based questions)

Write necessary code to solve the questions and perform all the tasks as per the grading rubric

Clearly write down observations, insights, and recommendations for the business problem based on the analysis performed

Once the notebook is complete, download it as a **.ipynb** file and convert it to a **.html** file

EasyVisa - Full-Code Version

The notebook should be submitted as an HTML file (.html) and NOT as a notebook file (.ipynb)

The conversion can be done via one of the following ways:

Jupyter Notebook: ***Download as HTML*** from the ***File*** menu
Google Colab: Use free online tools

Please make sure that all the sections mentioned in the grading rubric have been covered in the submission

EasyVisa - FAQ

I am getting the below error. How do I resolve it?

```
ValueError: could not convert string to float
```

As algorithms available in the sklearn library cannot handle non-numeric data, it is preferable to either encode or create dummy variables for the categorical features.

EasyVisa - FAQ

Tuning XGBoost is taking too much time. What to do?

Please find below the options that can be tried to reduce the run time for XGBoost tuning:

1. Pick individual features at a time and try adjusting them. Get the optimal value beyond which no improvement is seen, note that optimal value, and move on to the next feature.
2. Only use parameters that are necessary. Do not use all the parameters just because they are there.
3. Do not use more than 2-3 options in each list of hyperparameters given to the tuning function.
4. Try a small dataset randomly chosen from the original dataset, tune hyperparameters on them, and then use that set of hyperparameters over the entire dataset.
5. **The above steps can be followed to reduce the execution time while doing the hyperparameter tuning of other models as well.**

EasyVisa - FAQ

I am getting the below error with Bagging Classifier with Logistic Regression as base_estimator:

```
AttributeError: 'str' object has no attribute 'decode'
```

The `LogisticRegression`function has a parameter named solver, which was earlier by default set to 'liblinear', but after an update in sklearn, the default solver was changed to 'lbfgs'.

Kindly try setting the solver as liblinear, and try the code again, as shown below:

```
bagging_lr = BaggingClassifier(base_estimator =
LogisticRegression(solver='liblinear', random_state=1),random_state=1)

bagging_lr.fit(X_train,y_train)
```

EasyVisa - FAQ

I am getting the below warning while fitting XGBoost Classifier

```
WARNING: Starting in XGBoost 1.3.0, the default evaluation metric used
with the objective 'binary:logistic' was changed from 'error' to
'logloss'. Explicitly set eval_metric if you'd like to restore the old
behavior
```

To remove the warning kindly try setting the `eval_metric` hyperparameter as 'logloss', as shown below:

```
xgb = XGBClassifier(eval_metric='logloss')
```

Power Ahead!